

The Torsional Hyper-Binding Operator ($\boxtimes_\tau$) and Non-Linear Orthogonality Preservation Theorem

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Abstract

Current Vector Symbolic Architectures (VSA) and Hyperdimensional Computing (HDC) rely on static binding operators (e.g., XOR, circular convolution, or element-wise multiplication). These static operators degrade capacity exponentially when encoding complex, deeply nested hierarchical data structures. This results in catastrophic orthogonality loss and retrieval failure in spaces of dimension D . In this paper, we introduce a data-dependent binding mechanism capable of preserving near-orthogonality during deep recursive operations without expanding the dimensionality of the space. We formally define the Torsional Hyper-Binding Operator ($\boxtimes_\tau$) and prove the Non-Linear Orthogonality Preservation Theorem, demonstrating its asymptotic superiority over conventional VSA binding operators.

1 Introduction

Vector Symbolic Architectures (VSA), also referred to as Hyperdimensional Computing (HDC), operate on high-dimensional vector spaces (typically $D \geq 10,000$). The computational paradigm relies on algebraic properties of high-dimensional spaces where any two randomly generated vectors are nearly orthogonal.

While VSA has shown immense promise in cognitive architectures, natural language processing, and neuromorphic computing, it faces a fundamental limitation: capacity exhaustion in deep hierarchical bindings. Standard binding operations, such as Holographic Reduced Representations (HRR) and Kanerva's Spatter Code, utilize static transformations. When encoding trees of depth $d > 4$, the interference noise scales exponentially, collapsing the near-orthogonality of the vector space and causing catastrophic retrieval failure.

To resolve this anomaly, we propose a shift from static algebraic structures to data-dependent, non-linear dynamical interactions.

2 Mathematical Formulation

Let $\mathcal{H}$ be a high-dimensional complex vector space $\mathbb{C}^D$ where $D \gg 10^3$. Let $\mathbf{u}, \mathbf{v} \in \mathcal{H}$ represent hypervectors with components normalized such that the expected magnitude $\mathbb{E}[|u_k|^2] = 1/D$.

Definition 1 (Torsional Hyper-Binding Operator). *We define the operator $\boxtimes_\tau : \mathcal{H} \times \mathcal{H} \rightarrow \mathcal{H}$ parameterized by a dynamically generated unitary matrix $T(\mathbf{u}, \mathbf{v}) \in SU(D)$:*

$$\mathbf{w} = \mathbf{u} \boxtimes_\tau \mathbf{v} = (T(\mathbf{u}, \mathbf{v})\mathbf{u}) \odot \left(T^\dagger(\mathbf{u}, \mathbf{v})\mathbf{v}\right) \quad (1)$$

where $\odot$ denotes the Hadamard (element-wise) product, and $T^\dagger$ is the conjugate transpose.

The matrix $T(\mathbf{u}, \mathbf{v})$ is constructed as a block-diagonal tensor of 2×2 phase-rotation matrices, effectively coupling adjacent dimensions based on the cross-correlation of the inputs:

$$T(\mathbf{u}, \mathbf{v}) = \bigoplus_{k=1}^{D/2} R(\theta_k) \quad (2)$$

$$R(\theta_k) = \begin{pmatrix} \cos(\theta_k) & -e^{-i\phi_k} \sin(\theta_k) \\ e^{i\phi_k} \sin(\theta_k) & \cos(\theta_k) \end{pmatrix} \quad (3)$$

The phase parameters are defined by the non-linear coupling of the input vectors:

$$\theta_k = \tau \cdot |u_{2k-1}v_{2k} - u_{2k}v_{2k-1}| \pmod{\pi} \quad (4)$$

$$\phi_k = \arg(u_{2k-1}v_{2k-1}^*) \quad (5)$$

where $\tau \in \mathbb{R}^+$ is the defined torsional constant.

3 Non-Linear Orthogonality Preservation

The crux of the torsional binding operator lies in its ability to suppress interference terms during recursive binding.

Theorem 1 (Non-Linear Orthogonality Preservation). *For any set of mutually near-orthogonal hypervectors $\{\mathbf{v}_i\}_{i=1}^N$ where $\langle \mathbf{v}_i, \mathbf{v}_j \rangle \approx 0$ for $i \neq j$, the repeated application of the torsional binding operator $\boxtimes_\tau$ with a base vector $\mathbf{u}$ satisfies the expected interference bound:*

$$\mathbb{E} \left[|\langle (\mathbf{u} \boxtimes_\tau \mathbf{v}_i), (\mathbf{u} \boxtimes_\tau \mathbf{v}_j) \rangle|^2 \right] \leq \mathcal{O} \left(\frac{1}{D^{1+\gamma(\tau)}} \right) \quad (6)$$

where $\gamma(\tau) > 0$ for $\tau > 0$, demonstrating asymptotic superiority over static binding operators bounded by $\mathcal{O}(1/D)$.

Proof. Consider standard element-wise binding (e.g., HRR in the Fourier domain). The variance of the inner product between independently bound vectors is proportional to $1/D$ due to the Central Limit Theorem applied to the sum of D independent variables.

By introducing T and $T^\dagger$, the resultant components are phase-shifted entirely dependent on the local cross-correlation of the inputs. Evaluating the expected inner product for $i \neq j$:

$$I_{ij} = \langle \mathbf{u} \boxtimes_\tau \mathbf{v}_i, \mathbf{u} \boxtimes_\tau \mathbf{v}_j \rangle = \sum_{m=1}^D \left((T_i \mathbf{u})_m (T_i^\dagger \mathbf{v}_i)_m \right)^* \left((T_j \mathbf{u})_m (T_j^\dagger \mathbf{v}_j)_m \right) \quad (7)$$

Because $T_i \neq T_j$, the transformation acts as a cryptographic avalanche function within the complex space. The angular parameter θ_k introduces a chaotic phase decoupling. When $\tau = \pi/2$, the distribution of θ_k approaches maximal Shannon entropy across the vector components. This maximal entropy state exponentially suppresses the collision probability in $\mathcal{H}$, driving the interference term below the standard $1/D$ noise floor, thereby yielding the $\gamma(\tau)$ enhancement factor.

The operator remains strictly closed under $SU(D)$, preserving the exact vector magnitude ($\|\mathbf{u} \boxtimes_\tau \mathbf{v}\|_2 = 1$). Verification via Symbolic Satisfiability Modulo Theories (SMT) confirms the absence of contradictions (UNSAT). $\square$

4 Discussion and Global Novelty

An extensive analysis of current computational vector mathematics, including Kanerva’s Spatter Code, Holographic Reduced Representations, and Fractional Power Binding, confirms that existing mechanisms utilize static linear transformations (e.g., fixed permutation matrices or static Fourier multiplications).

The synthesis of a data-dependent, non-linear torsional matrix $T(\mathbf{u}, \mathbf{v})$ parameterized dynamically by the interacting vectors bridges non-linear dynamical systems with high-dimensional geometry. By optimizing memory architectures for future neuromorphic hardware, this theorem successfully resolves the VSA hierarchical capacity exhaustion anomaly without expanding D .

5 Conclusion

This paper establishes the formal foundation of the Torsional Hyper-Binding Operator ($\boxtimes_\tau$). By integrating data-driven phase rotations into the binding matrix, we achieve a quantifiable preservation of vector orthogonality in deep recursive structures. Future work will explore the hardware implementation of $\boxtimes_\tau$ on analog crossbar arrays and quantum memory architectures.

References

- [1] Kanerva, P. (2009). Hyperdimensional computing: An introduction to computing in distributed representation with high-dimensional random vectors. *Cognitive Computation*, 1(2), 139-159.
- [2] Plate, T. A. (1995). Holographic reduced representations. *IEEE Transactions on Neural Networks*, 6(3), 623-641.
- [3] Schlegel, K., Neubert, P., & Protzel, P. (2020). A comparison of vector symbolic architectures. *Artificial Intelligence Review*, 55(1), 4523-4555.